

Grid Path Maximum Minimum Score

You are given a grid.
Each cell contains a value.

Aadya wants to go from $(0, 0)$ to $(n-1, m-1)$ moving:

Up

Down

Left

Right

Score of a Path

Suppose path is:

$8 \rightarrow 5 \rightarrow 7 \rightarrow 4$

Score becomes:

$\min(8, 5, 7, 4) = 4$

because the smallest value encountered determines the path's score.

Question

Return the path whose score is maximum.

Solution Code

```
class Solution {
public:
    int maximumMinimumPath(vector<vector<int>>& grid) {

        int n=grid.size();
        int m=grid[0].size();

        vector<pair<int,int>> dir={{-1,0},{1,0},{0,-1},{0,1}};

        //step 1 prepare maxheap for {score,{row,col}}
        priority_queue<
        pair<int,pair<int,int>>> pq;

        //we need a array to score max score to each cell
        vector<vector<int>> score(n,vector<int>(m,-1));

        pq.push({grid[0][0],{0,0}});
        score[0][0]=grid[0][0];

        while(!pq.empty()){

            int currscore=pq.top().first;
            int r=pq.top().second.first;
            int c=pq.top().second.second;

            pq.pop();

            if(currscore<score[r][c]) continue;
            if(r==n-1 && c==m-1){
                return currscore;
            }

            for(auto it:dir){

                int nr=r+it.first;
                int nc=c+it.second;

                if(nr>=0 && nr<n && nc>=0 && nc<m){

                    int nextscore=min(currscore,grid[nr][nc]);

                    if(nextscore>score[nr][nc]){

                        score[nr][nc]=nextscore;
                    }
                }
            }
        }
    }
};
```

```
                pq.push({nextscore,{nr,nc}});  
            }  
        }  
    }  
    return -1;  
}
```